



Area S: central aisle of the basilica, looking east.

and less uniformly on the outer sides of the walls facing the eastern and western aisles. The central aisle is divided into three tiers, separated by two ashlar steps across the aisle. Flanking each step is a pair of pilasters. Of the southern pair, only the carved base remains, but the northern pair survives to a height of 1.5–2 m; their fallen carved capitals were found at their base. Two stones with a rough boss are incorporated into the eastern pilaster; the bosses are partly hidden in the wall and the stones are probably in secondary use.

The northernmost tier seems to have been the focus of the building. The row of corbels in the northern wall indicates that this space was roofed differently than the rest of the building. Aside from these plain corbels, another 12 finely dressed corbels were found in the collapsed debris, apparently having supported the roof above the central aisle. Similar finely worked corbels were found in the mansion in area K and in the synagogue.

The finds in the building are meager and provide no evidence concerning its original function. In the eastern aisle were traces of fire, ashes, and charred wood, which, together with the great quantity of large iron nails found on the floor, indicate that a large piece of furniture may have stood there. Next to the northeastern pilaster were traces of a hearth and a complete storage jar, which are certainly out of place in a public building, and are evidence of the use of the building during an emergency. It was apparently converted for use by refugees during the siege in 67 CE. Finds in a similar context have been discovered in the synagogue and in the public square above the olive-oil press in area R. Fragments of several knife-shaped oil lamps and limestone vessels were found in the fill and debris covering the building. Noteworthy are a fragment of a mold-made lamp, dated to the mid-first century CE; and a narrow, lathe-turned hard limestone vessel, which may be an inkwell. In the central aisle a seventh specimen of the locally minted autonomous “Gamala” coins was discovered, as well as an apparently unique find in Israel—a bronze coin of Amphipolis in Macedonia, dated to the second century BCE.

The size and height of the building, the opulence of the central aisle, the size of the stones of the central aisle’s walls, the width of the doorways,

the corbels, and the pilasters leave no doubt as to the public character of building 2100. Each of its aisles is considerably larger than any other room at Gamala except the synagogue, and in the setting of the Golan it is highly unlikely to have been a private residence, even of a very wealthy person. The wide aisles, especially the western one, necessitated very long roof beams, which must have been imported. Most of the building was apparently plastered. The main entrance was no doubt in the southern wall, opening onto the central aisle, and there may have been entrances opening onto each of the side aisles. It is possible that the decorated lintel found in area S by S. Gutman was set above the main entrance.

A tentative suggestion for the architectural origin of the building can be offered. If the two walls of the central aisle, pierced by two or three wide doorways on each side, are regarded as two rows of piers, they lend the building the appearance of a Roman basilica with piers instead of columns.

The building dates to the early first century CE. It is not a synagogue, as it contains no single area that is suitable for the assembly of a large group of people. Rather, it seems to be an entirely new type of Jewish public building of the Second Temple period—whether religious, secular, or both. Given the paucity of archaeological data on public buildings of this period, caution is required in categorizing the building architecturally, ascertaining its function, or attempting to associate it with functions mentioned in the rabbinical sources. If, as is believed, the synagogue was the only Jewish communal center in Talmudic times and provided all necessary services, then the corollary of the discovery of this building is that in the Second Temple period the situation was different, that there was at least one function that was separate from those of the synagogue, and that it took place in a building with its own architectural form. The subject requires further study.

DANNY SYON, ZVI YAVOR

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GERIZIM, MOUNT

An update to Vol. 2, pp. 484–491.

HISTORY

Archaeological finds—pottery, coins, and radiocarbon dating—indicate that the temple on Mount Gerizim was first built in the mid-fifth century BCE. A city began to grow up around it at the end of the fourth century BCE, after the city of Samaria was destroyed by Alexander the Great. At the beginning of the second century BCE, in the days of Antiochus III, the temple and its compound were rebuilt and the city expanded greatly. Based on the coins found at the site, it is clear that the temple and the city were destroyed by John Hyrcanus around 110 BCE, and not in 128 BCE as recounted by Josephus, who writes that immediately after the death of Antiochus VII Sidetes (128 BCE), Hyrcanus launched a campaign against Mount Gerizim (*Antiq.* XIII, 254–257; *War* I, 62).

The city and temple stood abandoned through the Hasmonean and Roman periods, during which the Samaritans made a number of unsuc-

cessful attempts to reestablish their control of the mountain (*Antiq.* XVIII, 85–89; *War* III, 307–315). During the reign of Antiochus Pius (160 CE), the upper temple of Zeus on the northern ridge of Mount Gerizim was inaugurated, and at the beginning of the third century CE, the temple was completed by one of the Severan Dynasty. At the beginning of the fourth century, with the rise of Christianity, the Samaritans reconstructed the sacred precinct and converted it into a pilgrimage site. In the Late Roman period, a fortress was erected near the precinct. In 484 CE, Zeno erected a fortified church to Mary Mother of God on the ruins of the sacred precinct. The church was reconstructed and expanded by Justinian after it was damaged in the Samaritan revolts. During the Crusader period the Samaritans successfully returned to Mount Gerizim to conduct sacrifices; a monumental Samaritan inscription was found from this period.



Mount Gerizim: plan of the site.



View of the sacred precinct, looking west.



View of the sacred precinct, looking south.

EXCAVATIONS

Excavations at the site, under the direction of Y. Magen, began in 1983 and continued for 20 years until 2003. Large parts of the Hellenistic city and the Samaritan sacred precinct were uncovered in the excavations. The Byzantine period Church of Mary Mother of God and its compound on the southern ridge and the remains of the temple of Zeus to the north were excavated by R. Bull.

EXCAVATION RESULTS

The archaeological discoveries on Mount Gerizim can be divided into the Hellenistic city, the sacred precinct from the Persian period, the sacred precinct from the Hellenistic period, the Roman temple of Zeus, the remains to the north of the Roman temple, the renewed Samaritan cult on the mountain in the fourth century CE, and the Church of Mary Mother of God.

THE HELLENISTIC CITY. The Hellenistic city grew up around the sacred precinct of the Persian period, which was built on the highest spot of the southern range of Mount Gerizim, overlooking Shechem and the eastern valleys. Two major roads reached the city. One ascended from Shechem to the west through a pass to the Roman temple. The other, from the east, is the extension of the route to Shechem from Jerusalem in the south. The city founded on Mount Gerizim had no natural advantages, having been built on a high, barren mountaintop with no water sources, far from the central crossroads of Samaria. It was established for religious reasons at a holy site and functioned as a sanctuary and city of priests.

The city covered an area of 800 by 500 m—over 100 a. Most of the dwelling quarters were situated on its southern and western sides, where the rocky terrain slopes moderately. Few buildings stood on the northern side, and construction on the steep eastern slope required large-scale hewing into the bedrock, with very few buildings constructed there as well. The city was not entirely surrounded by a wall and its perimeter was not determined by walls and towers. No urban planning is apparent. It grew and expanded slowly with the addition of buildings, streets, and neighborhoods. During the Hasmonean campaign against the city, the lack of a defensive wall necessitated the construction of barricades and the individual fortification of streets and houses. The only truly fortified area in the city was the sacred precinct erected during the second construction stage in the Hellenistic period.

The city was divided into residential quarters: the southern quarter (area B), which was the largest, including the citadel to the south of the city (area G); the southwestern quarter (area A); the mansion (area P); the eastern slope (area E); the northern quarter (area T); and the northwestern quarter (area K).

The Southern Quarter (Areas B and G). The densely inhabited quarter at the southern edge of the city is surrounded by an improvised peripheral wall formed by the adjoining walls of the outer houses. The quarter measures c. 300 m from east to west and c. 120 m from north to south and was traversed by a central road which began at the two-chambered southern gate (8.50 by 7.50 m) and ended at two large squares. Alleyways leading to houses branched off to the east and west. On either side of the gate stood two sizeable public buildings, each constructed around a central courtyard. The western building measured 19.30 by 16.50 m; the eastern building measured 21 by 17 m and had two entrances, one from the north and the other directly from the city gate. At the southwestern end of the southern boundary wall formed by the outer walls of the houses protruded a large citadel measuring 25 by 25 m and constructed around a central courtyard. All the exterior openings of the gate structures and the citadel had been blocked, apparently at the time of the siege of the city. To the northeast of the city gate was uncovered another group of buildings, which included buildings B1 and B2. B1 comprised a central building surrounded by storerooms; B2 was a large building with a central courtyard and an elaborate reception room, its façade adorned by two Doric columns, forming a portico.

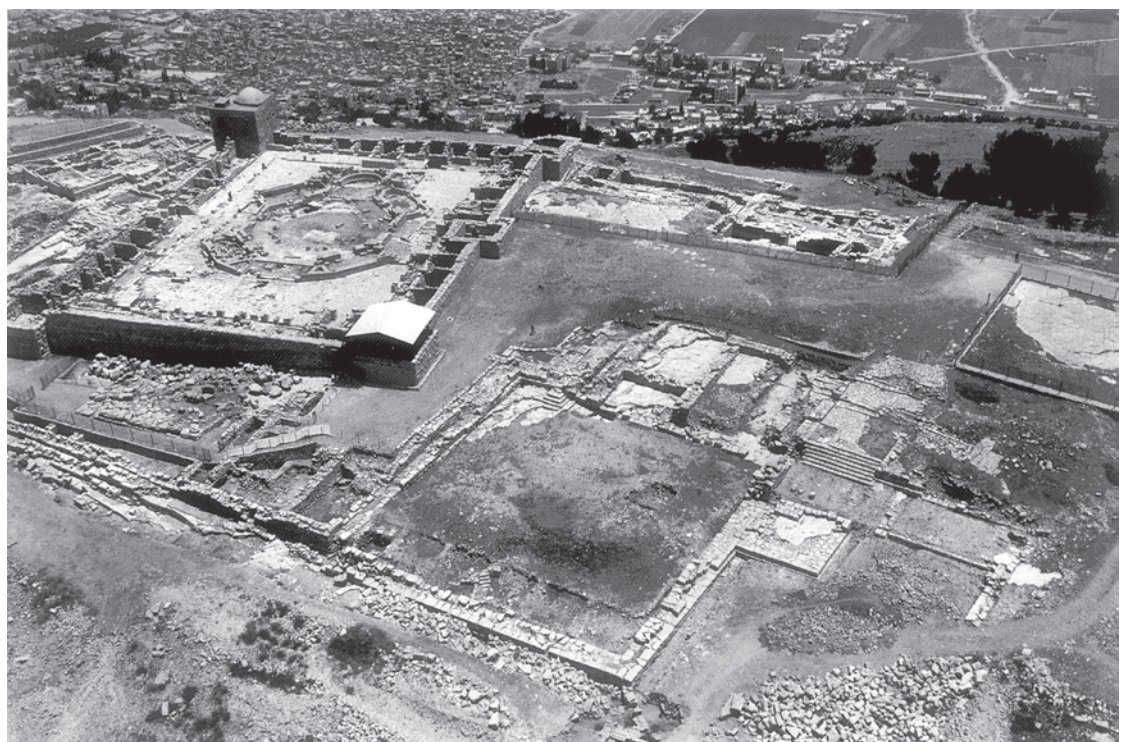
The southern section of the bound-

ary wall of the quarter was uncovered for 250 m. At its eastern end were enclosed courtyards. Some 180 m to the south of the boundary wall a fort (28 by 27 m) with four square towers (7 by 7 m) was revealed (area G), situated in a mountain pass along the road ascending to the city from the east. The finds inside the fort were few, but include coins from the end of the fourth century BCE up to the days of John Hyrcanus.

The Mansion (Area P). To the west of the sacred precinct, a large mansion was uncovered, unlike any other structure found as yet on Mount Gerizim. This architectural complex, built on a steep slope, was composed of three wings: an olive-oil press (building I), a dwelling and commercial section (buildings II–III, V), and luxurious living quarters (building IV). Measuring 40 by 40 m, the complex has five entrances—two in the north, one in the east, one in the west, and one in the south. It was constructed abutting the wall of the sacred precinct and in certain places has been preserved up to its second story.

Building I (22 by 19 m) was built around a paved court. The entrance, from the northern passage, had a well-hewn stone threshold into which was set a wooden door. In the large eastern hall a crushing installation and two niches for anchoring the beams of an olive-oil press were found, with weights still in situ. This building may originally have been a dwelling, which was converted into an olive-oil press at a later stage, similar to the building in area T (see below). To the west of the press was a luxurious dwelling, building IV, outstanding in the Hellenistic city on Mount Gerizim. It measured 20 by 14 m and was built around a stone-paved central courtyard with a cistern. The entrance was from the northern passage. This building had a magnificent reception room (13 by 5.5 m) with walls covered in stucco and colored frescoes, and a floor in white plaster. This is the only example of colored frescoes known so far on Mount Gerizim, suggesting that this building belonged to a wealthy individual. To the south of the olive-oil press was uncovered a group of buildings (II–III, V), including dwellings, storerooms, and shops that opened onto another passage to the south of the complex.

The Eastern Slope—“The Bakers’ Fort” (Area E). The eastern side of the city was characterized by steep slopes that hindered the construction of houses. As the Seleucid authorities did not permit fortification of the city, these steep cliffs could not be exploited for defensive purposes. Instead, large public buildings were erected to great heights, creating a wall of sorts. On the eastern slope to the south of the sacred precinct, two buildings were excavated, E1 and E2. The former was a large building, 42 by 25 m, apparently a public structure rather than a dwelling, given the absence of large cisterns. On its northern side a staircase descended eastward from the sacred precinct, measuring 8 m long and 2 m wide, and including 15 steps, all preserved intact. The staircase led into a vestibule which opened onto the central courtyard surrounded by dwelling rooms and a central reception hall on the western side. The building contained over 20 cooking and baking ovens and was therefore dubbed “The Bakers’ Fort.” E2 was a smaller building located on the upper eastern slope.



View of the sacred precinct, looking northeast.

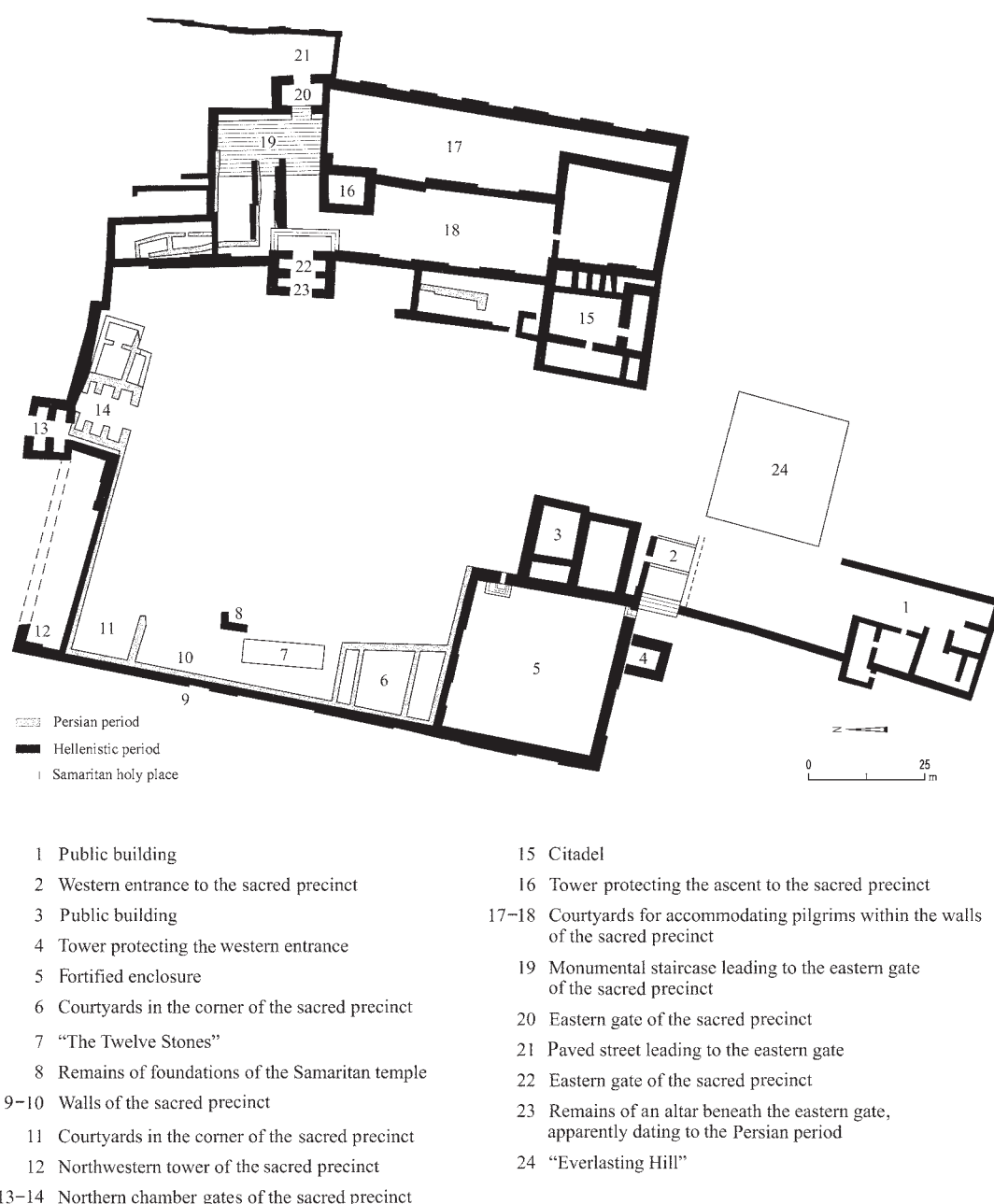
THE SACRED PRECINCT. The Samaritan sacred precinct, the first building erected on Mount Gerizim, was constructed in the mid-fifth century BCE (the Persian period) on the highest point of the mountain. In the second century BCE, at the beginning of the reign of Antiochus III, the temple and its enclosure were rebuilt following their destruction by John Hyrcanus.

The Persian Period Compound.

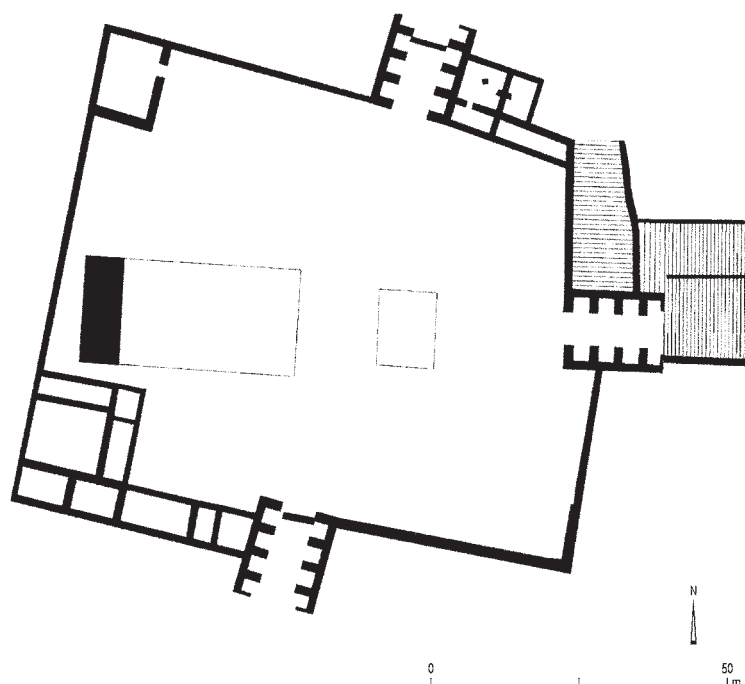
The dimensions of the Persian period compound were 98 by 96 m, including the gates. The 1.30-m-wide western wall is preserved to a height of 2 m along its entire length of 84 m. Despite the easy access to the site from the west, no gates were uncovered in this wall due to the location of the holy of holies at the rear of the temple (which can probably be identified with the present-day Samaritan holy site known as "the twelve stones"). The western wall was built of large fieldstones hewn from the upper layers of bedrock, as opposed to the stones of the later Hellenistic compound, which were hewn from deep quarries. Enclosed courtyards stood at either end of the western wall. The southern courtyard measured 21.5 by 12 m; the northern courtyard was 2.5 m wide, its length unknown. The northern compound wall was completely uncovered, measuring c. 73 m long and at its center a six-chambered gate, 15 by 14 m. To the east of the gate stood a large building (12 by 11 m) in which thousands of burnt bones of sacrificial animals were found in a thick ash layer. This building, located next to the altar, may have served as a receptacle for ash from burnt sacrifices. The eastern wall of the compound was extensively damaged by later building in the Hellenistic and Byzantine periods. The eastern gate of the Persian period may also be assumed to have been six-chambered, like the one in the north, with a staircase leading up to it. Many finds from the Persian and early Hellenistic periods were preserved in this spot. The southern gate was also almost completely destroyed by Hellenistic construction.

Thus, in the Persian period compound there were apparently three gates, on the northern, eastern and southern sides, reminiscent of the temple described in the Book of Ezekiel as a model for the Second Temple in Jerusalem. Thousands of pottery vessels and Persian period coins were found in the sacred precinct, providing the mid-fifth century BCE founding date for the temple. It and the surrounding compound continued in use for c. 250 years until the construction of the new temple in the Hellenistic period.

The Hellenistic Period Sacred Precinct. At the beginning of the second century BCE, a large, magnificent new compound was built in the sacred precinct, at its center a temple of white ashlars. It existed for some 90 years until its destruction by John Hyrcanus. (The precinct remained abandoned until the fifth century CE, when the construction of the Byzantine church and enclosure destroyed the Hellenistic compound to its foundations, with only its outer walls surviving.) In the Hellenistic period significant changes were made to the original plan of the precinct. For one, it was not modeled after the temple in Jerusalem, but followed a different style incorporating elements of Greek architecture. Although the core of the compound retained its original size, buildings and fortifications were added around it, increasing the outer circumference of the enclosure, which now measured 212 by 136 m. The western wall of the compound remained as it was, but a plastered offset-inset wall was adjoined to its outer face. The southern wing of the compound was drastically altered, with the addition of large public buildings and a monumental 9-m-wide staircase, seven steps



Plan of the Persian and Hellenistic period sacred precinct.



Hypothetical reconstruction of the Persian period sacred precinct.

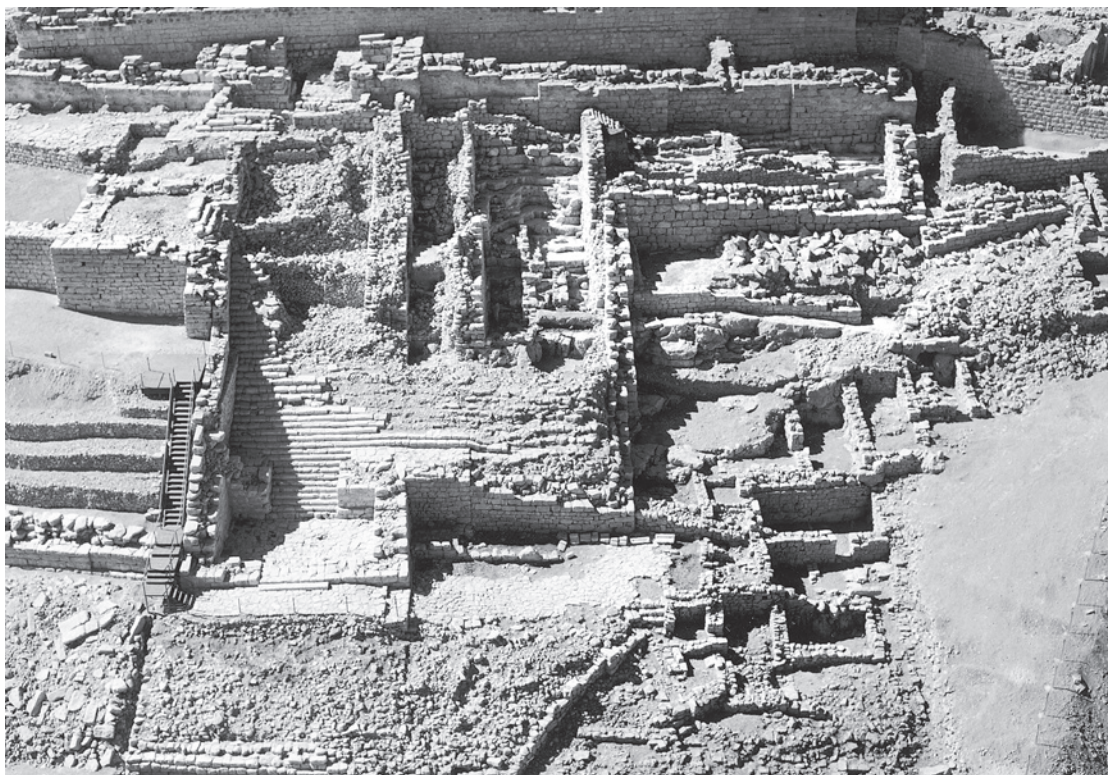
of which are preserved, changing the direction of the entrance to the compound. The stairs led to a paved square, from which one turned north toward the temple. Opposite the staircase was a square tower, 9 by 8.5 m, built to defend the gate. The entrance to the sacred precinct was also protected by a fortified enclosure, 39 by 38 m, with 3-m-thick offset-inset walls, entirely plastered. To the south were large public buildings connected to the temple and used by visitors. The northern wall of the compound was preserved for its entire length, together with the northern gate. The Hellenistic offset-inset wall adjoined the earlier Persian wall from the outside. In the northwestern corner stood a fortified tower built of large stones. The earlier, six-chambered northern gate was now replaced by a small but elaborate four-chambered gate. The entrance was 3.60 m wide and each chamber, measuring 3 by 2.5 m, was well paved in stone slabs. This gate was built to the north of the earlier gate to provide access to the compound during construction. It was 10 m long.

The eastern wall, 93 m long, was built on the steep slope. In the center was a large, four-chambered gate measuring 14 by 10 m with an entrance 4.7 m wide. The northeastern corner was built of especially large stones set deep into the ground. At the southeastern corner of the compound stood a large citadel, 25 by 24 m, with very thick walls rising to a second story, perhaps a third. The citadel was built around a central courtyard containing an elaborate and well-preserved reception room. The eastern wing of the compound was fortified by two thick, offset-inset walls forming large courtyards, in which pilgrims gathered prior to ascending to the temple.

The eastern gate served the pilgrims visiting the temple from Shechem. A paved street, up to 16 m wide, led southward to a two-chambered lower gate. From here, a staircase, up to 23 m wide and 34 m long, rose the 15.5 m to the upper levels by way of some 57 steps. It was enclosed on its northern and southern sides by thick walls and a monumental watchtower. At the top of the staircase was an elaborate, four-chambered upper gate with two wooden doors whose copper hinges were preserved in situ.

THE FINDS. Hundreds of thousands of finds were uncovered in the excavations at Mount Gerizim, including some 16,000 coins, pottery vessels, stone vessels, glass vessels, metal objects, inscriptions, and over 400,000 bones of sacrificed animals.

Coins. Sixty-eight coins were found from the Persian period, prior to the conquest of Alexander the Great, including Sidonian and Samaritan coins, most coming from the sacred precinct. The earliest coin is dated to



View of the monumental staircase leading to the eastern gate of the sacred precinct.

480 BCE. From the Ptolemaic period, 417 coins were found, including issues of Alexander the Great and Ptolemy I to VI. A *yhd*-type coin dating to 270–285 BCE was discovered. Most of the coins found on Mount Gerizim date to the Seleucid period, to the reigns of Seleucus II and III, and Antiochus II–VIII. Also found were many Acco coins. As mentioned above, Mount Gerizim was not destroyed in 128 BCE after the death of Antiochus IV, but in 110 BCE during the reign of Antiochus IX. From the Hasmonean period, 546 coins were recovered, dated to the reigns of John Hyrcanus, Aristobulus, and Alexander Jannaeus. There is a break in the numismatic record at Mount Gerizim from after Jannaeus until the beginning of the fourth century CE, but numerous Late Roman, Byzantine, Late Islamic, and Crusader period coins were retrieved.

Animal Bones. In the sacred precinct at Mount Gerizim, over 400,000 burnt bones of sheep, goats, cattle, and pigeons, most below the age of one year, were found within layers of ash. The main concentrations of burnt bones were found in the following locations: on the northeastern side of the compound near the northern gate, most of these dating to the Persian and Ptolemaic periods; outside and along the eastern wall of the compound; in the fortified enclosure in the southwestern corner, near the western staircase; and on the inner, southwestern side of the Persian compound, near the “twelve stones.”

Inscriptions. Recovered in the sacred precinct of Mount Gerizim were over 500 inscriptions, as well as masons’ marks on building stones, paving stones, and a sundial. Some 400 inscriptions in ancient Hebrew and Aramaic are dated to the Hellenistic period. Most of the names mentioned, in addition to the titles of Cohen and Levite, are biblical Hebrew names. Eighty-three of the inscriptions are in Greek, some from the Hellenistic period; but most Samaritan dedications to the temple on Mount Gerizim are from the fourth and fifth centuries CE, when the Samaritans restored the sacred precinct. A few are from the Byzantine church. An inscription written in Samaritan script can be dated to the Crusader period. Over 20 ashlar, apparently from the temple itself, bear Hebrew and Greek masons’ marks.

Weapons. Many weapons were found at the site, including lead sling projectiles, bronze arrowheads, and a complete iron sword.

EVIDENCE FOR A TEMPLE ON MOUNT GERIZIM. Many of the finds from the sacred precinct indicate the presence of a temple on Mount Gerizim during the Persian and Hellenistic periods. The excavations revealed dozens of well-cut ashlar, bearing masons’ marks, which had been removed from the walls of the temple. Its large stones were reused in the church, and proto-Ionic capitals belonging to the early temple of the Persian period were found. The inscriptions discovered on Mount Gerizim provide the strongest evidence for the existence of a temple there. These include, for example, the titles of the priests who served in the temple and the formula *qdm 'lh' b'tr' dn'* (before God in this place), indicative of the site’s sanctity. Another inscription includes the formula *qdm 'dny*



Inscription in Proto-Jewish script.

(before the Lord), and an additional clearly legible inscription in Hebrew reads "that which Joseph offered for his wife and his sons before the Lord in the temple." Another inscription mentions *byt dybh* ('house of sacrifice'), indicating a temple (2 Chr. 7:12). In an inscription in Neo-Hebrew script, the tetragrammaton *yhwh* appears, apparently originally reading "house of Yahweh." A Greek inscription on a sundial reads "God Most High." A fascinating discovery is a small gold bell with a silver clapper of a type originally attached to the fringes of the ephod of a high priest (Ex. 28:33–35).

THE SAMARITAN SACRED PRECINCT IN THE FOURTH AND FIFTH CENTURIES CE. Following the disappearance of paganism and the designation of Christianity as the official religion of the Roman Empire, a religious reform was initiated by the Samaritan leader Baba Rabah. The Samaritans took advantage of this time of transition to renew their cult on the sacred precinct on Mount Gerizim and construct many synagogues. Evidence for the existence of the Samaritan holy site on Mount Gerizim during the fourth century CE can be found in the Christian sources of the period and in Midrash (*Bereshit Rabah* 32:10). It is unknown whether there was only a synagogue for prayer or if animals were also sacrificed inside the Samaritan sacred precinct, as in the Hellenistic period. Dozens of Samaritan inscriptions in Greek have been found at the site. Most begin with the formula "the one God who helps" along with the Samaritan Hebrew names of those who dedicate gold coins to the holy site on Mount Gerizim.

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GESHER BENOT YA'AQOV

An update of Vol. 2, pp. 493–495.

INTRODUCTION

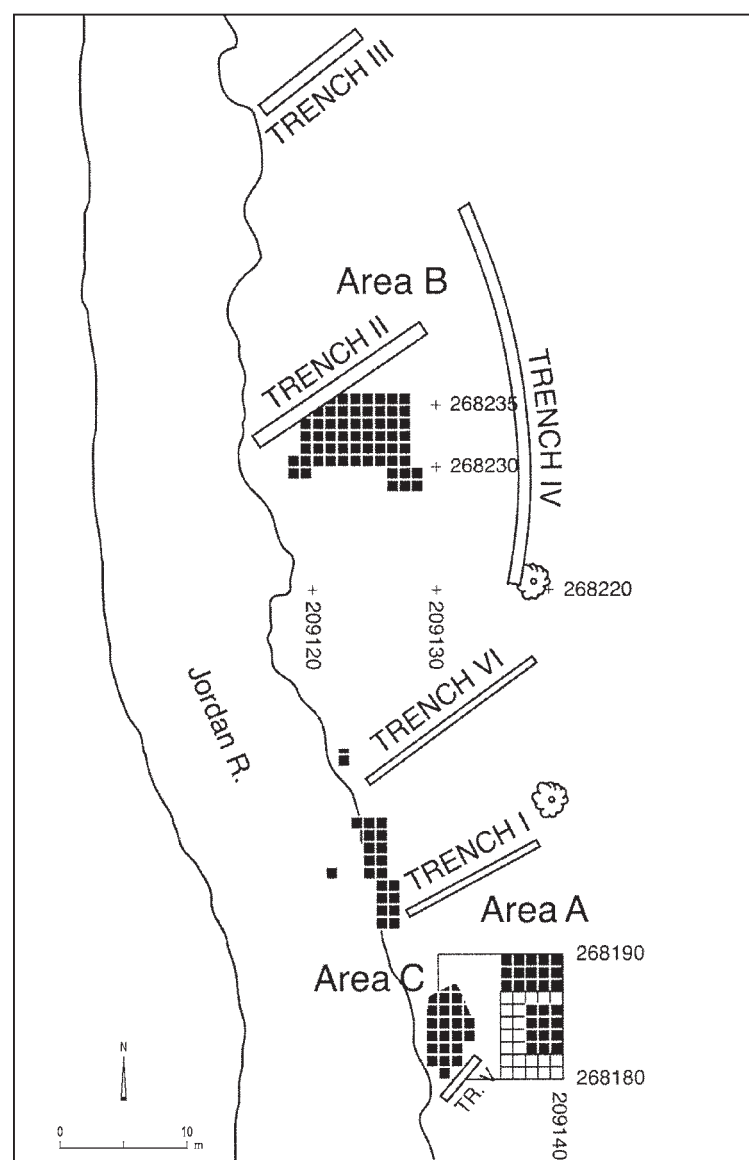
Between 1989 and 1997, seven field seasons were directed by N. Goren-Inbar of the Hebrew University of Jerusalem at Gesher Benot Ya'aqov to facilitate recognition of the stratigraphic sequence. The sections, representing a 34-m-thick stratigraphic sequence, revealed that the geological structure is tilted up to 45 degrees at the study area. Excavations adjacent to the trenches (in three major areas: A, B, C) exposed over 13 Acheulian archaeological horizons comprising a wealth of botanical and paleontological assemblages. The lithic assemblages retrieved reflect the typical Acheulian tool kit: hand axes, cleavers, flake tools, flakes, and cores.

A paleomagnetic study located the Matuyama/Brunhes chronological boundary (790,000 years ago) in the lower third sector of the sequence, thus assigning a greater antiquity to the formation than had previously been assumed. The estimated duration of the entire depositional sequence at the study area is c. 100,000 years, assigning it to Oxygen Isotope Stage 18–20, thus making Gesher Benot Ya'aqov one of the most unique terrestrial sequences known from this time range.

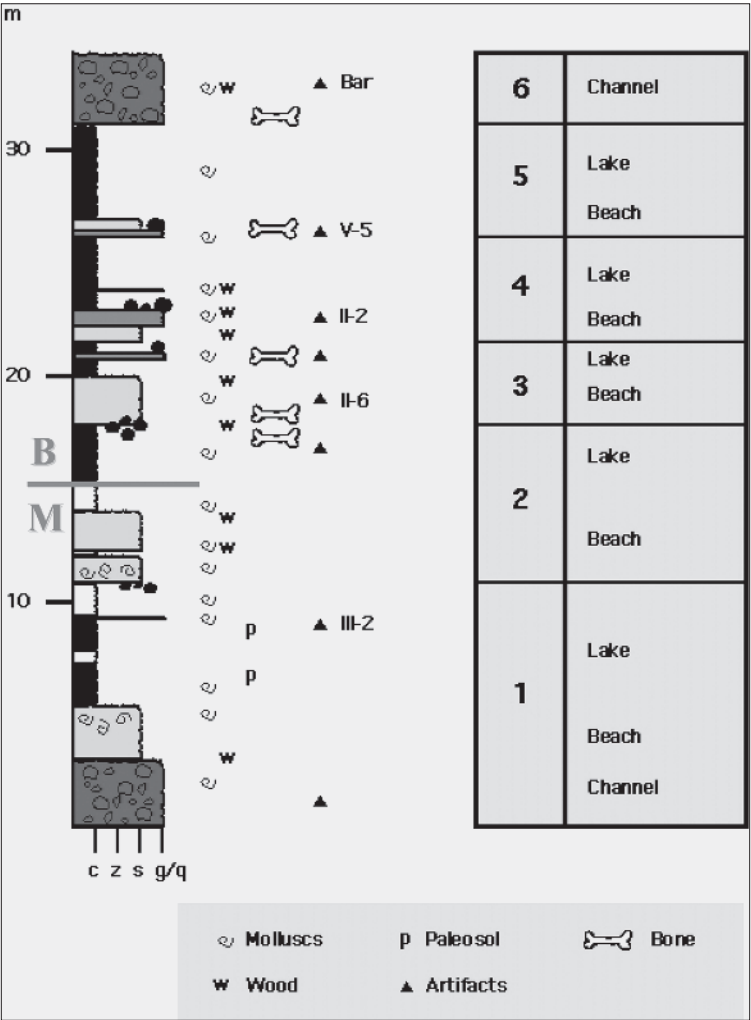
The site has been greatly influenced by the water regime of the paleo-Lake Hula. The water level fluctuated in a similar manner to that of the modern lake, causing the deposits to change from relatively deep-water sediments to very shallow, littoral facies. In addition to their inundation by the shallow lake, the waterlogged sediments were influenced by several fluvial systems that flowed into the rift from the Galilee Mountains in a general west–east direction (Naḥal Rosh Pinna, Naḥal Maḥanayim, Naḥal Dishon, and Naḥal Ayelet Hashaḥar). The stream loads from these systems were deposited in a series of fans that are frequently observed in the Hula drillings. The intensive morphotectonic changes in the region throughout the Pleistocene caused these fluvial systems to shift their courses in adjustment to the newly formed topographies. These developments are responsible for the waterlogged nature of the Gesher Benot Ya'aqov site. In addition, the formation of the Jordan River, a young phenomenon in the region, was an important component in preserving the waterlogged nature of the Early–Middle Pleistocene sediments. These conditions ensured the outstanding preservation of organic material in the sediments of Gesher Benot Ya'aqov. The archaeobotanical research of this material is in turn a crucial component of the attempt to reconstruct the paleoenvironment of Lake Hula and its environs.

EXCAVATION RESULTS

THE PALEOBOTANICAL EVIDENCE. Rich botanical assemblages retrieved from the excavations include wood, bark, fruits, and seeds. Within the wood finds, 27 genera of trees, shrubs, and woody climbers were identified and are the primary data set for a basic understanding of the paleobotany of the Dead Sea Rift zone and its adjacent areas. The set



Gesher Benot Ya'aqov: trenches and excavation areas.



Composite section and sedimentary cycles.

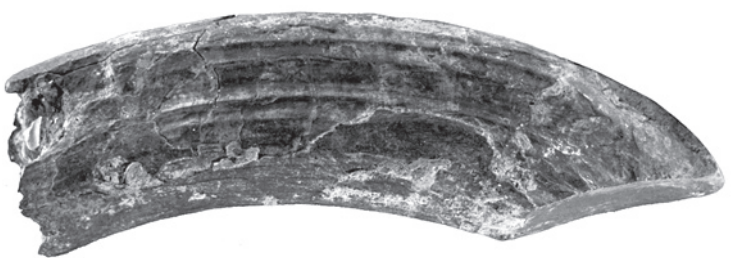
also indicates that this assemblage was not affected by the continued presence, during hundreds of thousands of years, of hunter-gatherer groups that populated these parts of the eastern Mediterranean region.

In addition, over 130 taxa of fruits and seeds were identified, forming the most extensive botanical database of the Levantine Lower Paleolithic. An important aspect of the botanical finds is the edible component of these assemblages. An example is the 7 species of edible nuts: five extant terrestrial nuts and two extinct aquatic nuts that were found associated archaeologically with pitted hammers and anvils. Another botanical aspect of the site is a single worked wood item—a man-made polished wooden plank of a willow tree (*Salix*) (25 by 13.5 by 4 cm), the earliest known item of its kind in the world.

THE PALEONTOLOGICAL EVIDENCE. Extensive paleontological assemblages were recovered and include mammals (elephants, rhinos, cervids, bovids, hippos, and equids) and micro mammals. Also found are birds, fish, crabs, and an especially large collection of molluscs. Many



Area C: general view.



Hippo/potamus tusk.

of the species identified are important paleontological landmarks from both the perspective of biogeographical aspects and chrono-evolutionary significance.

THE LITHIC ASSEMBLAGES. The bifaces (hand axes and cleavers) of Gesher Benot Ya'eqov have been found in all archaeological horizons excavated at the site. The most characteristic tool type of the Acheulian material culture, they are mainly made on basalt flakes whose width generally exceeds their length. These flakes have pronounced, frequently thinned bulbs of percussion and striking platforms, and their axes of symmetry are either oblique or perpendicular to the axes of blow. The technological properties are observable due to the fact that later modification by retouch has been minimal and did not mask the earlier technological features. A detailed study demonstrates that the bifaces have been modified on flakes produced by several different techniques. It seems that hand axes and cleavers cannot be distinguished from one another on the basis of their size or shape properties. Raw material for the manufacture of these tools was probably procured from the site's immediate vicinity, as inferred from the presence of several basalt giant cores within the finds retrieved from layer II-6. All these cores show large flake scars and some clearly display features indicative of the prepared core technique. The initial stage of biface manufacture consisted of producing large flakes of lenticular cross-section and a thin distal working edge. The second stage was flake modification, which consisted of the removal of extra thickness as well as minimal edge treatment. This reduction process is unique in terms of its uniformity and the homogeneity of the end products, which attest to a considerable degree of standardization.

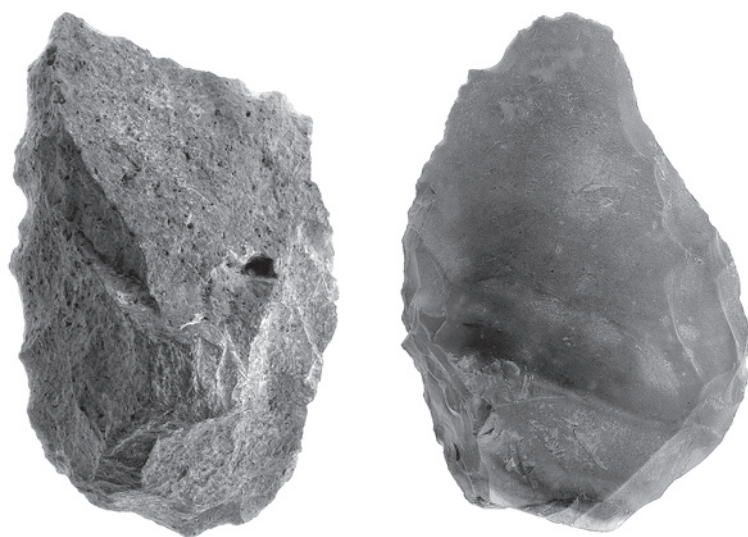
The study of the Gesher Benot Ya'eqov hominins' technological abilities also resulted in the discovery of the earliest recorded Levantine use of soft percussion mode. The evidence is observable on both the basalt and flint artifacts. Excavations in area C resulted in rich flint assemblages (particularly of layers V-5 and V-6) exhibiting this technique.

PATTERNS OF HOMININ BEHAVIOR. The behavioral pattern that emerges is a varied one, expressed by patterns of raw material exploitation, high functional diversity of the archaeological horizons, and exploitation of the Mediterranean flora and fauna in the vicinity of the site—a unique ecological niche of high potential and great diversity. Hominins had the cognitive abilities that enabled foresight (multi-stage planning) followed by realization of their plans. A specific knowledge of the animals' anatomy can be deduced from the location of cut marks and other types of bone damage and fractures caused by exploitation for marrow extraction and consumption.

Analysis of the lithic assemblages shed light on the complexity of behavioral patterns in the production of bifacial tools. It seems that different modes of activity took place: some bifaces were brought as pre-



Area B: elephant skull and underlying wooden log in association with basalt giant cores and other artifacts from layer II-6, level 1.



Basalt cleaver and flint hand axe on modified flake (ventral view).

forms to the site, others were introduced as finished objects, and in some cases the entire reduction sequence took place in situ. Of particular importance are the latest discoveries, which demonstrate the presence of controlled fire in the archaeological horizons, the earliest record in Eurasia. Evidence comes from burned flint microartifacts and burned wood. The mastery of fire is of significance for diet and protection. It offered outstanding opportunities and indicates an evolutionary advance unknown elsewhere in Eurasia.

NAAMA GOREN-INBAR

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GIV'AT HA-ORANIM

THE SITE

The Chalcolithic site of Giv'at ha-Oranim lies on a rocky hill, 130 m above sea level at its highest point, 15 km northeast of Ben Gurion Airport and 3 km east of Shoham, on the route of the Trans-Israel Highway (Route 6). The hill slopes moderately on all sides but the southern, which drops sharply toward Nahal Beit Arif. The densest occupation remains are concentrated on the southern and eastern parts of the hill. The site forms part of a larger site known as Khirbet el-Bira, named after a small Byzantine/Early Islamic period village excavated by Z. Safrai and S. Dar from 1980–1983.

EXCAVATIONS AND SURVEYS

I. Shavit of the Israel Antiquities Authority surveyed the site in 1994 and described the remains of small quarries, two basins, a rock-cut winepress, and agricultural terraces from the Byzantine period. No mention was made of Chalcolithic remains. In October 1995, a second survey was conducted under the direction of E. Oren and N. Scheftelowitz of the Institute of Archaeology, Tel Aviv University. With the exception of a single body sherd, no Chalcolithic remains were encountered. In 1996–1997, two excavation seasons were conducted. An area of c. 2.5 a. within the c. 100-m-wide, north–south strip of the planned road was excavated.

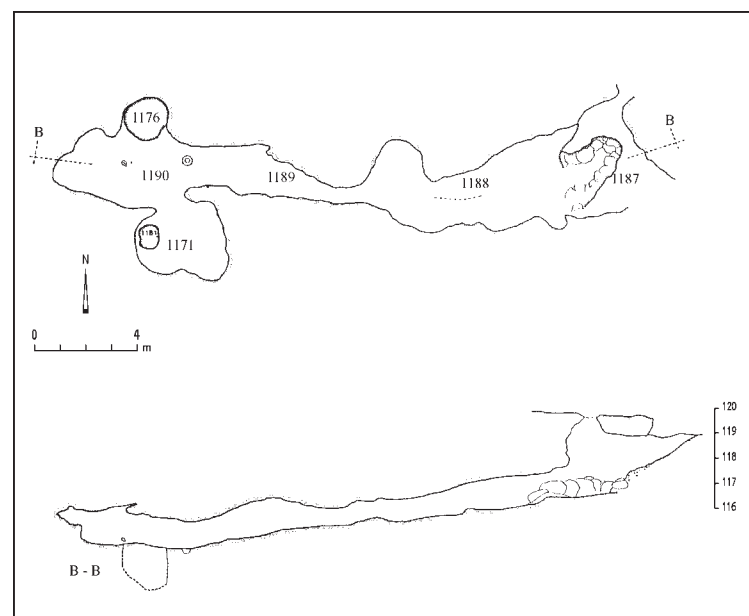
EXCAVATION RESULTS

Discovered in the 1996–1997 seasons were underground complexes including caves, shafts, pits, and graves from the Chalcolithic period, as well as Byzantine agricultural installations, walls, and an arcosolium tomb.

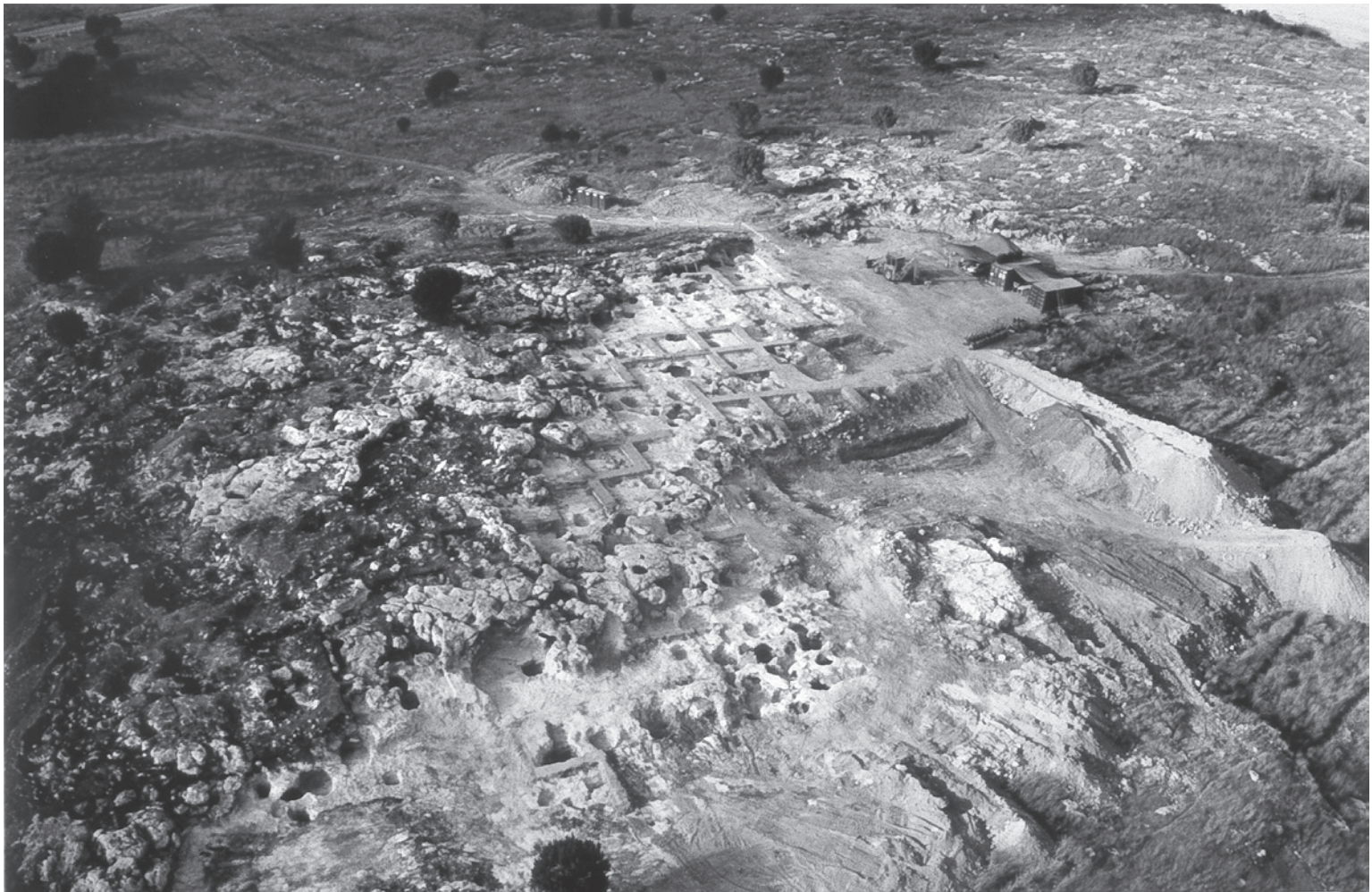
THE UNDERGROUND COMPLEXES. Nine multi-chambered caves were found at Giv'at ha-Oranim. These were natural karstic caves enlarged and adapted to human needs for living, storage, and burial. They are in a bedrock ridge that extends c. 200 m along the hilltop from northeast to southwest. Seven would have been suitable for habitation. The complexes consist of round rooms of various sizes connected by hewn openings or corridors. Access to some of these complexes was via hewn openings in their roofs. In most, the leveled bedrock was utilized as the floor, however some have evidence of compacted lime floors. Niches and pits were dug into the floors. The finds in the caves include a large quantity of complete ceramic vessels and sherds, basalt implements, flint tools, metal artifacts, and other small finds, together with human and animal bones.

The two uninhabitable caves each had a single entrance from the side. Cave 1185 is c. 27 m long and comprises four spaces: a vestibule, an outer room, a corridor, and an inner room. Finds include pottery, basalt implements, and human bones. Cave 1186 consists of a central space and an inner niche. Pits containing storage jars were found in both caves.

PITS. Pits excavated in the bedrock are the most common feature at the site. These were mostly scattered in areas within and outside the courtyards of the habitable caves. They are found as separate features or as components of a pit complex of two to more than ten interconnected pits. All were dug into the bedrock. Only in one case was a pit lined with stone. They may be divided into five main types: (1) bell-shaped—characterized by a flat bedrock floor, rounded walls, and a mouth of smaller diameter



Cave 1185: plan and section.



Giv'at Ha-Oranim: aerial view of the site.

than the floor; (2) barrel-shaped—identical to the first group except that the rim is of the same diameter as the floor; one pit of this type was stone-lined; (3) deep pits (or shafts)—similar to those of the first group, but over 3 m deep; only three pits of this type were found at the site; (4) shallow pits—less than 40 cm deep; and (5) irregularly shaped pits. Most of the pits were filled with earth, stones, pottery sherds, and other small finds, reflecting their final stage of use as refuse pits. They may have originally been utilized for refuse disposal, or for storage, working crafts, or cooking.

BURIALS. Human bones representing 15 individuals were found in 15 archaeological features in the caves and pits. A few bones were recovered in most of these, however there were several cases in which only a

single bone was found. The state of preservation was very poor and burial practices were verifiable in only three examples. Consequently, no comprehensive study was undertaken and the results are generally inconclusive regarding the age or sex of the individuals. In several burials the deceased were laid in a contracted position and were generally accompanied by pottery, basalt implements, flint, bone tools, and animal bones.

INSTALLATIONS. Several rock-cut installations were found in the western part of the site and in the bedrock near caves 1185 and 1186, as well as above some of the caves and on the eastern side of the site. There are several types of installations, including cup marks and cup mark series, basins, and surfaces for crushing and pressing (grapes or olives).

SMALL FINDS. Small finds include pottery, groundstone, and copper objects, in addition to bone tools, spindle whorls, loom weights, and two small rectangular mother-of-pearl pendant plaques, perforated on the upper part.

Pottery. The ceramic assemblage consists of storage, cooking, and serving vessels of various types and sizes: pithoi, storage jars, hole-mouth jars, bowls (mostly V-shaped in different sizes), churns, spoons, goblets, and fenestrated bowls. The vessels were decorated with paint, usually red, in bands or geometric patterns; some were adorned with impressions, incisions, or a plastic decoration. The majority of the pottery assemblage from the site is dated to the late Chalcolithic period and may be classified as Ghassulian ware.

Groundstone. A large assemblage of stone vessels was found at Giv'at ha-Oranim. Limestone is readily available in the immediate vicinity of the site. A total of 85 whole and fragmentary limestone vessels were found, including 18 bowls, 4 mortars, 8 grinding stones, and 7 querns. Several hundred whole and fragmentary basalt bowls and pedestal bowls were retrieved. Most are V-shaped bowls with straight or flaring walls, rounded or pointed rims, and thick bases of 4–7 cm. Pedestal bowls occur in small and medium sizes, and have three or four triangular legs. Sub-types include a bowl with three legs with a thick band between the stand and the base that is decorated with circular incisions; and a square pedestal bowl, c. 0.5 m high, with two pairs of loop handles at its sides. The basalt vessels are highly polished, most fragments plain while a few have incised herringbone patterns or a triangular decoration on the inner side of the rim. There is no evidence for any manufacture of basalt vessels at the site. They arrived to the site as finished products, imported from areas some 100–



View of underground complex.

200 km away. Other stone objects include mace heads, some made of haematite and manganese-oxides, pierced stones, and hammer stones.

Copper Objects. Twenty-four copper objects were found. These include three stands, two crowns, one "scraper," four mace heads, three axes, three adzes, and two pins. They are similar to items found in the Cave of the Treasure in Nahal Mishmar. The finds here include prestige objects as well as functional tools, along with six lumps of alloyed copper. The copper objects were found together with stone objects, bone tools, and pottery in several small separate caches.

SUMMARY

Giv'at ha-Oranim belongs to a group of contemporary Chalcolithic sites located within a 3 km radius (Shoham, Qula, and Tinsmet Cave) recently added to a list of sites excavated in the region beginning in the 1940s (Lod, Wadi Rabah, and Nahal Qaneh as well as burial sites located south of the Yarkon River). These sites attest to intensive settlement in this region during the course of the Chalcolithic period. Their material culture reflects a variety of influences and links with the Ghassulian culture. During the Byzantine period, the site was utilized for agricultural activities by the inhabitants of Khirbet el-Bira.



Basalt pedestal bowl.



Copper objects.

NAAMA SCHEFTELOWITZ

Main publication: N. Scheftelowitz & R. Oren et al., *Giv'at Ha-Oranim: A Chalcolithic Site* (Salvage Excavation Reports 1), Tel Aviv 2004.

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